Fixed work and wall-clock results

With up to four attempts per method, weight-aware First-Fit averaged 67.17% and the hybrid averaged 69.26%. Mean elapsed times were 7.4 ms and 144.9 ms. An attempt is a restart for First-Fit and a pool build plus a search for the hybrid. The attempt cap makes scores repeatable and does not equate computational work.

Budget or cap	Cases	First-Fit	Hybrid	Mean attempts FF / H
Up to 4 attempts	30	67.17%	69.26%	3.5 / 3.7
150 ms	30	69.11%	68.94%	258.9 / 3.1
500 ms	30	69.78%	70.25%	863.1 / 11.0
1.5 s	30	70.32%	70.69%	2,603.6 / 32.5
5 s subset	9	71.82%	72.14%	7,833.9 / 100.2

The sweep reused the 30 held-out cases with one trial seed. The 5-second row covers three cases per family. Both methods can stop early when they reach an upper bound. The earlier three-trial study starts on the next page.

Payload and mixed cases

Weight-only knapsack followed by First-Fit reached the 28.02% payload bound on all ten payload cases. That control averaged 70.26% on mixed cases. On the 30 original mixed pools, HiGHS proved a mean optimum of 85.62%, versus 83.41% for the recorded hybrid. The weight-and-volume bound averaged 98.59% and is loose because it ignores geometry.

What the hybrid samples

The production hybrid is a conflict-free compound search. Its conflict penalties are zero on every visited state, and it minimizes the slack bits analytically. It uses the reduced overweight penalty. A separate neal run sampled the explicit QUBO, including placement and slack bits. neal returned 0 valid raw layouts in 192 reads across 6 pools. The best repaired layout scored below the recorded compound search on every tested pool.

Coverage and support

On three geometry probes, changing corner priority raised the 20-box pool optimum from 91.25% to 100%. The other two already reached 100%. A larger cap and more layouts had no further effect on these probes.

Full-support utilization (First-Fit / hybrid) Geometry 98.14%/91.33% | Payload 27.71%/27.99% | Mixed 78.03%/79.58%

Post-repair support cleanup does not refill freed capacity. These are synthetic cases and the family-level intervals come from ten instances each. The sweep is a follow-up on the same cases.

TECHNICAL REPORT / 01

Hybrid Algorithm-Based 3D Container Loading Optimisation

Final technical report · Held-out experiment and fixed-pool diagnosis

Research question

How much space can two box-packing methods use when they get the same time? The project compares multi-start First-Fit with a hybrid that builds possible placements and uses an annealing search to choose among them.

Main result

Across 30 held-out instances at 150 ms per method, multi-start First-Fit averaged 69.41% utilization and the hybrid averaged 68.89%. The hybrid was 0.52 percentage points behind. The paired 95% interval was -1.89 to +0.61 points. The interval includes zero, so the experiment does not establish an overall winner at this budget.

The hybrid averaged higher utilization on payload and mixed cases. First-Fit averaged higher utilization on geometry cases. The exact fixed-pool check found that missing placements accounted for most of the hybrid’s geometry shortfall.

Problem and rules

Maximize the volume of boxes loaded into one container. Each box has fixed dimensions, weight and a rotation rule. A box can appear once, must fit inside the container, and cannot overlap another box. Total weight must stay within the payload limit. Face contact is allowed. Utilization is loaded volume divided by internal container volume.

Scope

The original rules cover geometry and payload. A separate full-support mode requires every elevated box to rest on the full area of its base. The model does not test load-bearing strength, stability, weight distribution or cargo securing.

Methods and test design

Multi-start First-Fit

Try one largest-volume-first layout, then try longest-edge order and seeded shuffled orders. On restarts the method also changes the order of allowed rotations. It keeps the highest-volume valid result completed before the deadline.

Candidate pool and hybrid objective

Build independent layouts from several package orders. Use the origin and exposed corners from each layout as placement positions, then combine the resulting choices up to a cap of 160. Each binary variable selects one placement.

The QUBO subtracts the volume of every chosen placement and adds a penalty for every selected overlapping pair or pair for the same package. A squared residual between total weight plus binary slack and capacity adds a payload penalty. The classical search minimizes over slack analytically at each move. Weight units use the greatest common divisor of the package weights and capacity, preserving original gram values.

A compound move inserts one placement and removes its blockers together. The sampler estimates a temperature range from move costs and finishes with a local improvement pass. Repair first removes conflicts, then uses exact knapsack selection if the remaining set exceeds capacity. The shared validator checks the final packing.

Held-out protocol

The frozen set has ten geometry, ten payload and ten mixed instances with 10–20 boxes. Geometry instances have saved full-volume tilings. Each method receives 150 ms per trial, including its preparation, search and validation. Late attempts are discarded. Three trials are averaged per instance. Paired bootstrap intervals resample the 30 instance differences.

Results at the same time budget

Family	First-Fit	Hybrid	Gap (pp)	W/T/L
All	69.41%	68.89%	-0.52	13/9/8
Geometry	98.72%	95.24%	-3.48	1/3/6
Payload	27.71%	28.02%	+0.31	4/6/0
Mixed	81.80%	83.41%	+1.61	8/0/2

Gap is hybrid minus First-Fit in percentage points. W/T/L counts hybrid wins, ties and losses after averaging each instance’s three trials. The aggregate interval crosses zero. Family results describe these generated cases at this budget.

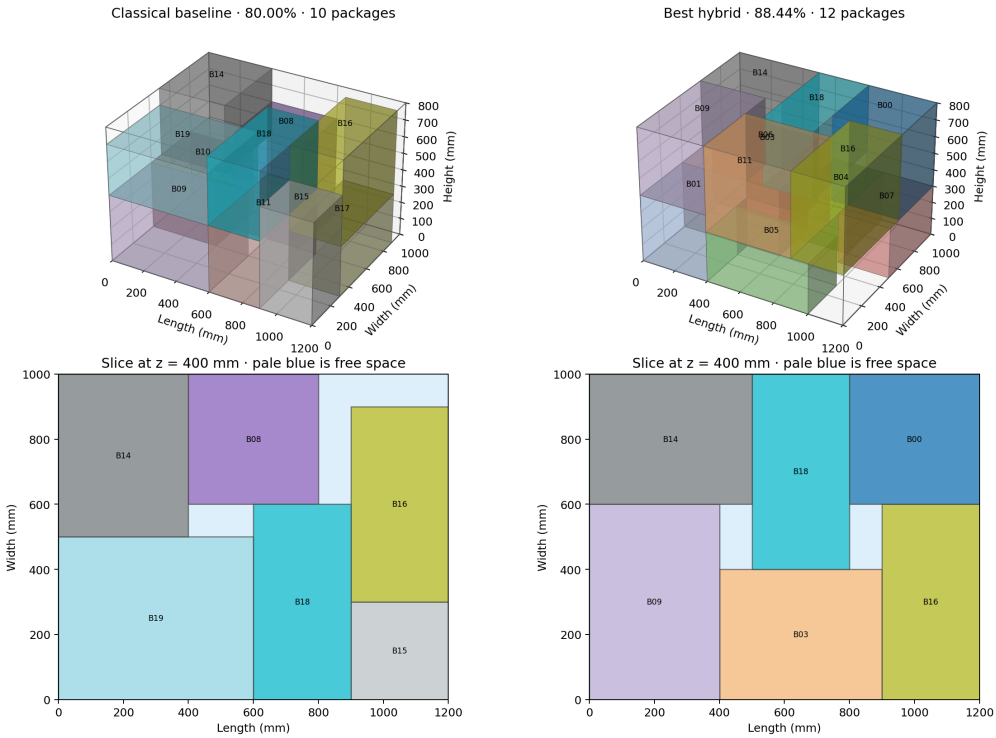


Figure 1. Recorded mixed case 4302, trial 301. First-Fit is on the left and the hybrid is on the right. The blue regions in the lower panels show free space at the same height. Panel scores are for this trial. Table scores are averages.

Validation

All 540 returned layouts across the brief, post-repair support and support-aware comparisons passed their respective validation rules. The benchmark includes cases where the first deterministic pass packed every box.

Original 150 ms mean attempts (First-Fit / hybrid) All 260.7/3.0 | Geometry 17.0/2.6 | Payload 668.4/3.7 | Mixed 96.6/2.8

What limits the hybrid

Candidate coverage explains most of the geometry gap

HiGHS optimized all 30 recorded geometry pools under the original rules. All optima were proven and 0 runs were time-limited. The pool optimum averaged 95.45%, against a 100% full-instance optimum. The hybrid reached the pool optimum on 27 pools. The mean gap was 4.55 points from candidate coverage and 0.21 points from search. 17 pools could not represent a full tiling.

Full base support

Hybrid support check	First-Fit	Hybrid	Gap (pp)
Post-repair check	67.96%	63.72%	-4.24
During search	67.96%	66.30%	-1.66

Full support requires the base of every elevated box to be covered by boxes directly below it. Checking support during search improved hybrid utilization over removing unsupported boxes afterward. All 90 returned raw support-aware results passed. The paired interval for the support-aware gap was -3.65 to +0.01 points.

Conclusion and next test

Multi-start First-Fit is a strong default at this budget. The hybrid merits further testing where payload and geometry compete. Improve candidate coverage before spending more time on the geometry search, then compare both methods across several fixed deadlines and new instance families.

The benchmark uses small synthetic instances and one machine. Equal deadlines can allow different numbers of completed attempts. The exact solver diagnoses recorded candidate pools and was outside the 150 ms comparison.

Appendix and reproduction

Development check with exact weight units

The tight 15-box development case returned to 100.00% mean utilization after removing 10 kg rounding. The 20-box case averaged 58.98% and reached 61.12% at best. All 45 exact-weight raw selections passed validation.

IBM Quantum fixture checks

On IBM ibm_fez, the expected bit pattern appeared in 1,013 of 1,024 shots. The three small QAOA packing fixtures had exact reference answers.

Fixture	Runs	Raw exact-optimum hit rate
H1	5	31.00%
H2	5	24.96%
H3	5	1.54%

Each run used 1,024 shots. These frozen fixtures verify circuit execution and measurement decoding. Their results are separate from the held-out comparison.

Reproduce and inspect

Start in container_loading and follow the README. Run heldout_benchmark.py for the main and support-aware comparisons, payload_experiment.py for the exact-weight development check, and geometry_pool_check.py for the fixed-pool diagnosis.

Evidence files include results/heldout/heldout.json, results/heldout/geometry_pool_check.csv, results/support_search/heldout.json, results/exact_slack/ablation.csv and results/hardware.json.